

Carrier Action

❑ Drift

Definition, drift current, mobility, resistivity

❑ Diffusion

Definition, diffusion current, Einstein relationship

❑ Recombination-Generation

Definition, R-G statistics, minority carrier lifetimes

❑ Equation of States

Continuity equations, minority carrier diffusion equations

❑ Supplemental Concepts

Diffusion lengths, quasi-Fermi-levels

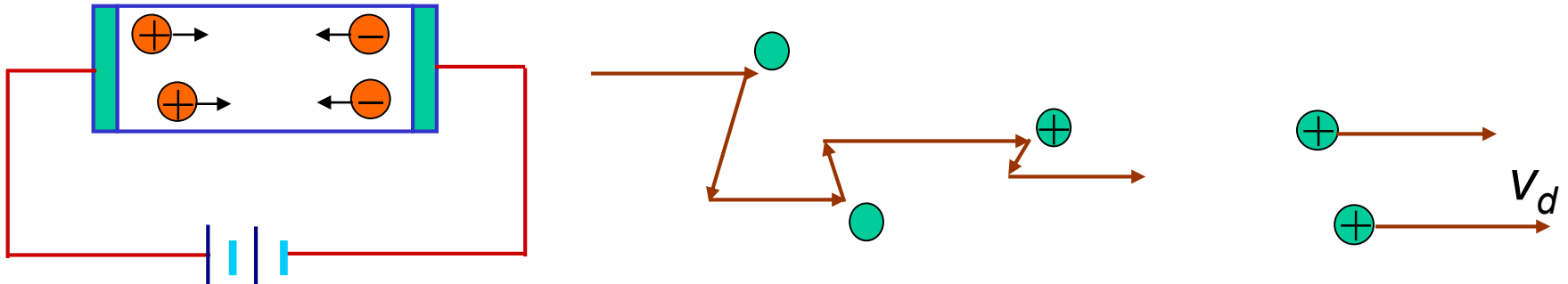
Drift: Definition and Physical Picture

□ Definition:

- Drift is charged-particle motion in response to an applied electric field

□ Physical Picture

- Microscope View
 - ✓ Charged particles get energy from E-field and move
 - ✓ Collide with other particles (atom, etc), change direction, and at same time lose energy
 - ✓ Again get energy from E-field and move, then repeat
- Macroscopic View
 - ✓ Viewed as a charged particle under electrical field



Drift: Drift Current

□ Drift Current

Consider the p-type semiconductor bar of cross-sectional area A

$v_d t$... all holes this distance back from the v_d -normal plane will cross the plane in a time t

$v_d t A$... all holes in this volume will cross the plane in a time t

$p v_d t A$... holes crossing the plane in a time t

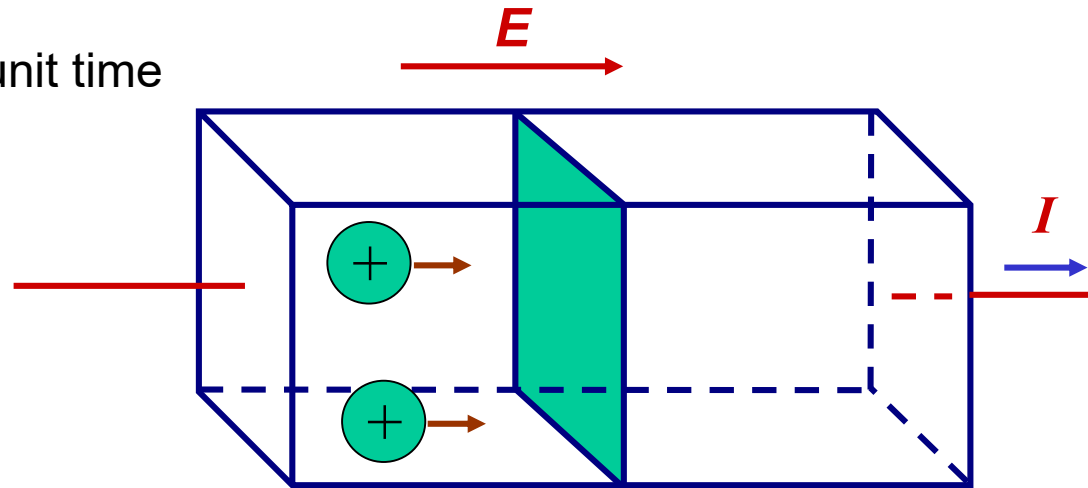
$q p v_d t A$... charge crossing the plane in a time t

$q p v_d A$... charge crossing the plane per unit time

Thus,

$I_{p,drift} = q p v_d A$... hole drift current

$J_{p,drift} = q p v_d$... hole drift current density



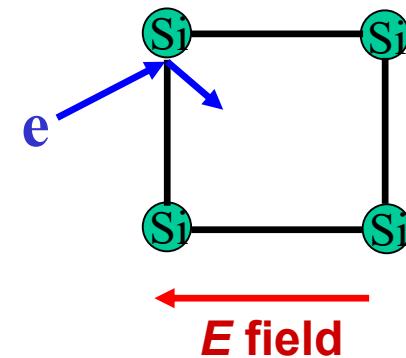
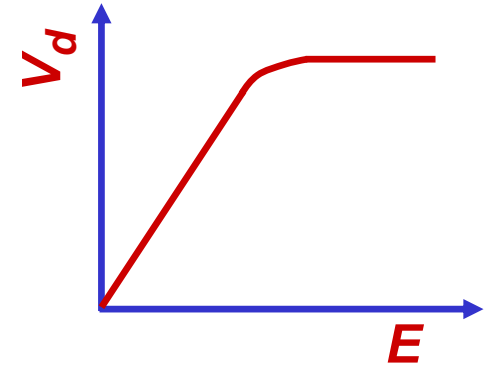
Drift: Drift Current with Electric Field

□ Drift Current (related with electric field)

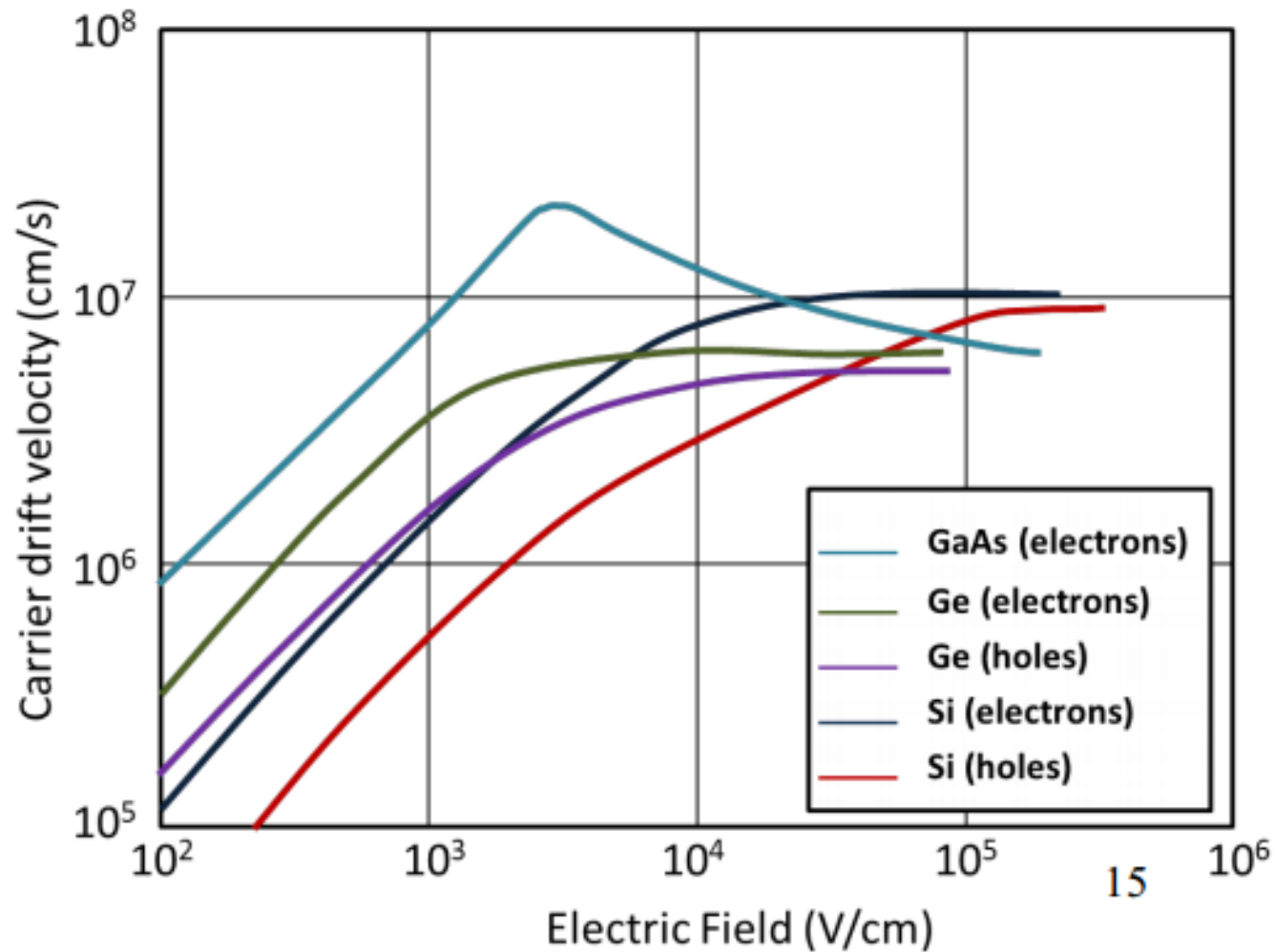
- From definition, drift current arises in response to an applied electric field, thus we try to relate the v_d with electric field
 - ✓ From experiments, v_d vs E relations
 - ✓ $v_d = \mu_0 E$, for $E < E_{sat}$ (μ_0 is mobility)
 - ✓ $v_d = V_{sat}$, for $E > E_{sat}$ (velocity saturation)
- Physical explain:
 - ✓ Why saturation (?)
- Thus, the drift current is given by

$$J_{p,drift} = qp v_d = q \mu_p p E$$

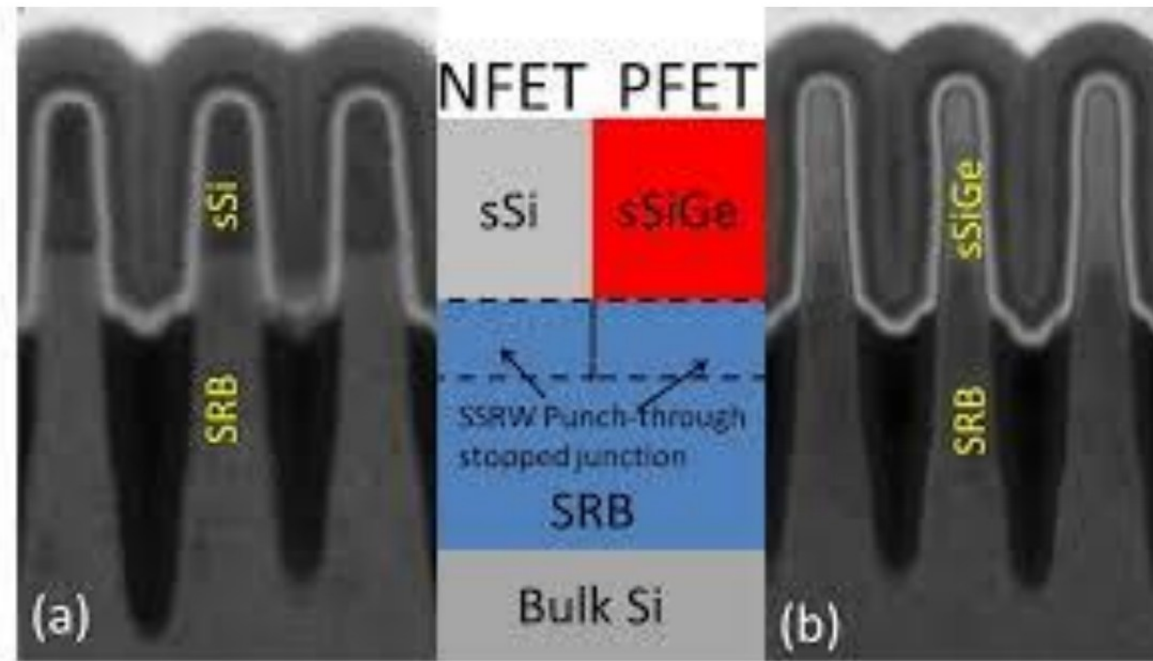
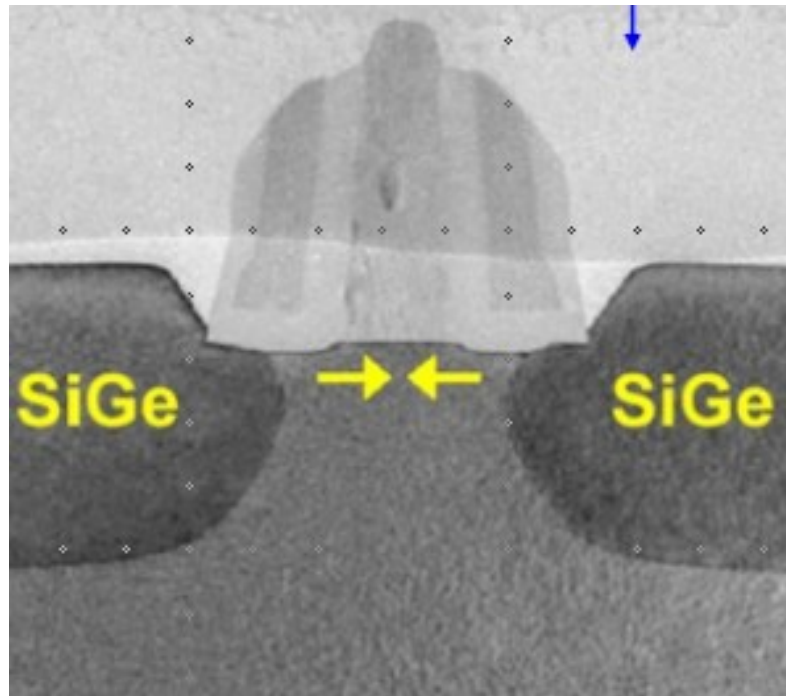
- ✓ $\mu_0 \rightarrow \mu_p$ for hole mobility
- ✓ only valid for low-field
- ✓ at high-field ($E > E_{max}$), then $J_{p,drift} = qp v_{sat}$, one factor limiting speed of MOSFET



Case Study of Drift Velocity



Some Important Applications of Germanium (Ge) in MOSFETs

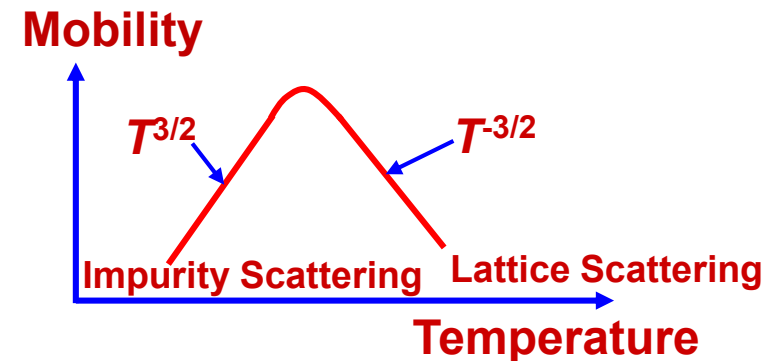
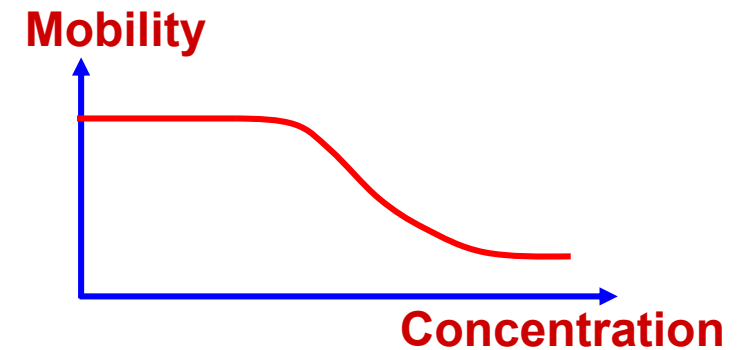


Drift: Mobility

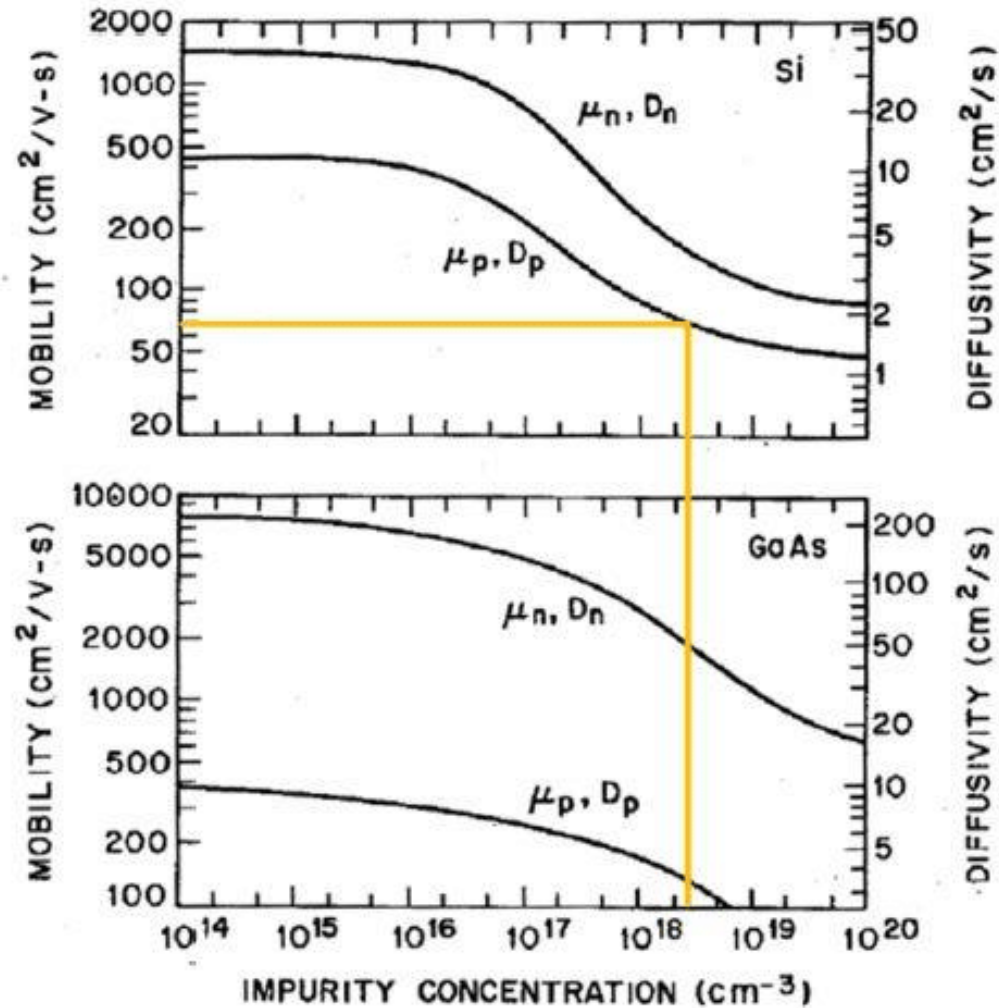
□ Mobility (μ)

- Definition ... is a key parameter in characterizing electron and hole transport due to drift ($\mu = q \langle \tau \rangle / m^*$)
- Physical view: mobility is a measure of the ease of carrier motion in a crystal
- Dominant Mechanisms:
 - ✓ Lattice scattering, involving collisions with heated lattice atoms
 - $\Rightarrow$ mobility increases with decreasing T
 - $\Rightarrow$ no doping dependence
 - ✓ Ionized impurity scattering
 - $\Rightarrow$ mobility decreases with increasing N_A or N_D
 - $\Rightarrow$ mobility increases with increasing T
- Consider all contributions of different scattering:

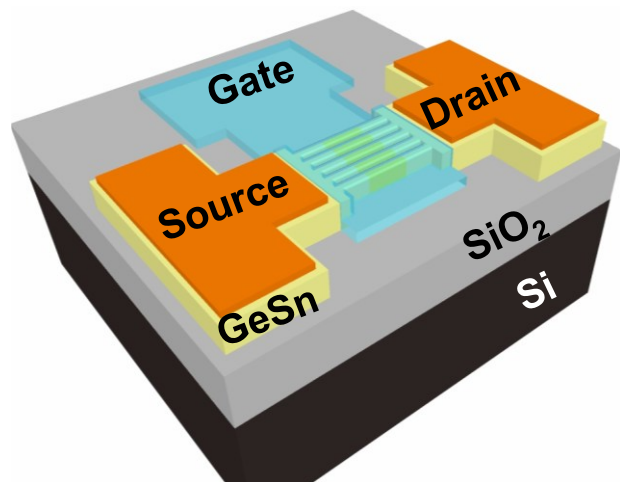
$$\boxed{\frac{1}{\mu} = \frac{1}{\mu_1} + \frac{1}{\mu_2} + \frac{1}{\mu_3} + \dots} \quad (\text{Mathiessen's rule})$$



Case Study of Mobility

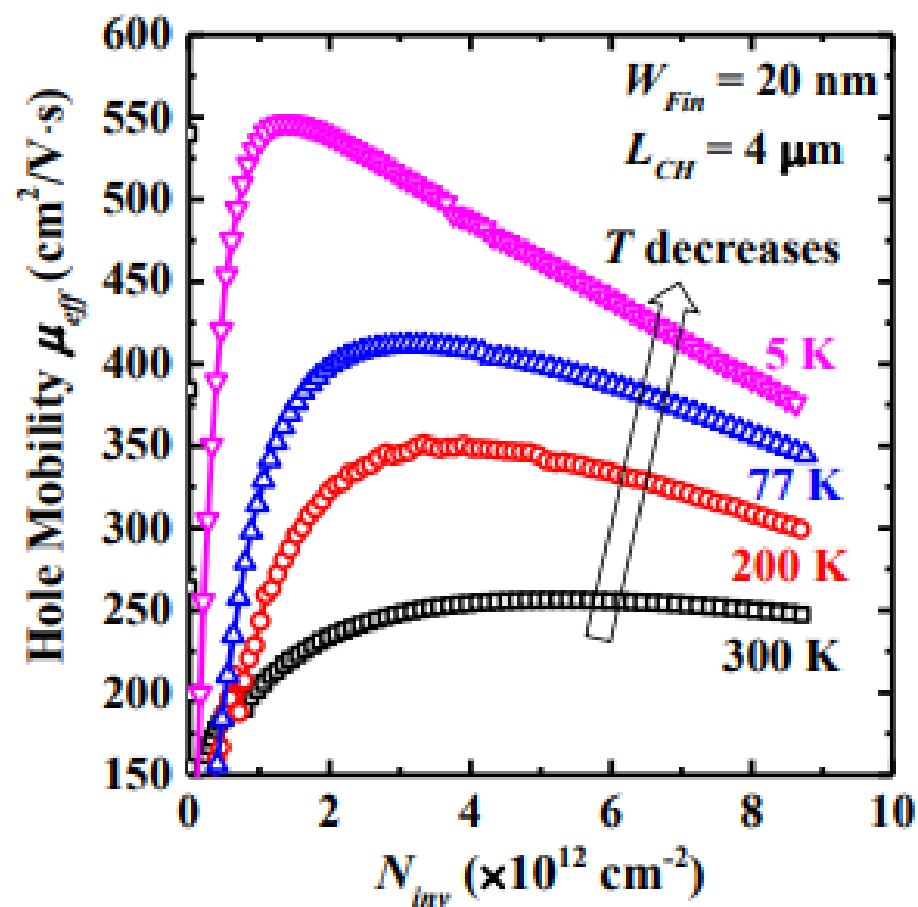
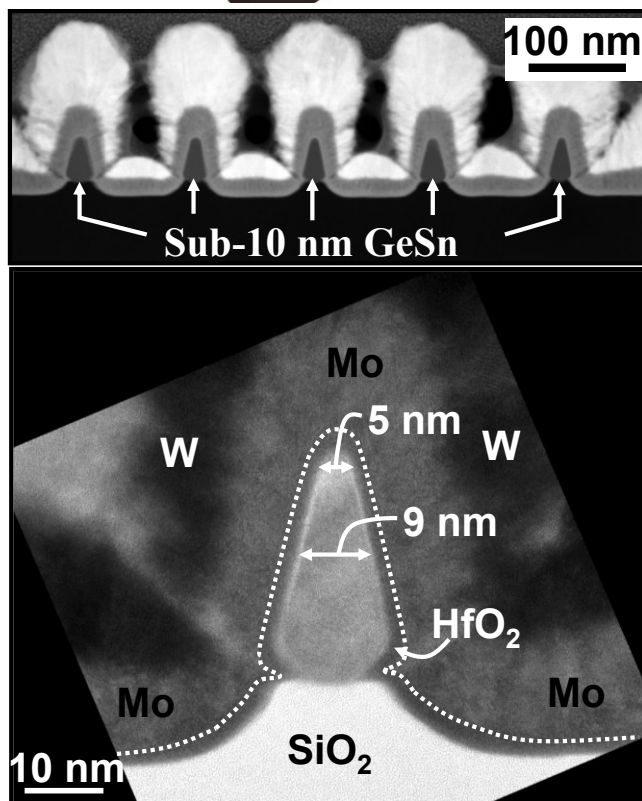


Case Study of MOSFET



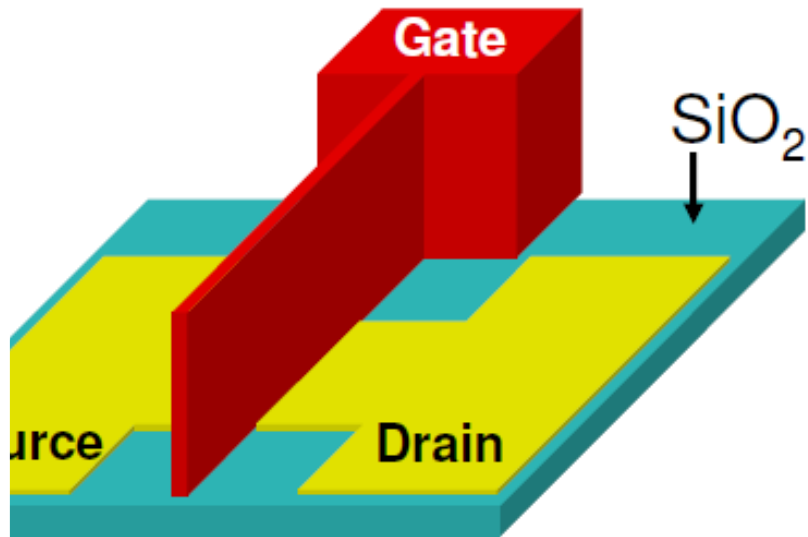
Group IV in Periodic Table

⁶ C
¹⁴ Si
³² Ge
⁵⁰ Sn

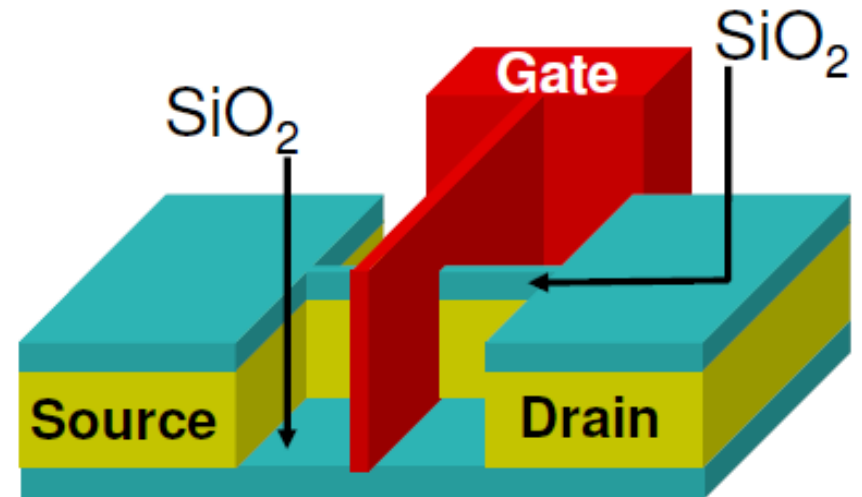


A brief introduction of FinFET

Bulk Planar Transistor



FinFET



- Simple process
- CMOS compatible

Drift: Resistivity

- Definition: resistivity is a measure of material's inherent resistance to current flow

$$J = \sigma E = E / \rho$$

$$J_{drift} = J_{p,drift} + J_{n,drift} = q(n\mu_n + p\mu_p)E$$

Thus, we have

$$\rho = \frac{1}{q(n\mu_n + p\mu_p)}$$

- Comments:

- For non-degenerated semiconductor with $N_D \gg n_i$

$$\rho = \frac{1}{qN_D\mu_n}$$

- For non-degenerated semiconductor with $N_A \gg n_i$

$$\rho = \frac{1}{qN_A\mu_p}$$

- Measure resistivity to determine dopant concentrations

Diffusion: Definition and Physical Picture

□ Definition:

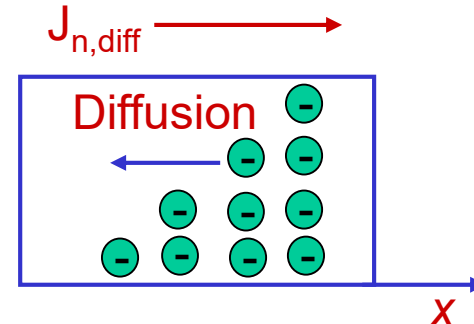
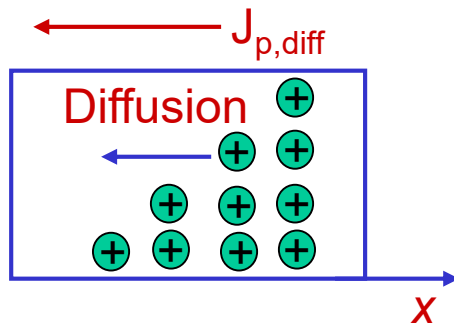
- Diffusion is particle motion result from its spatial variation

□ Analogy:

- Example “perfume” in a closed room

□ Physical Picture:

- Microscopic view: redistribute as a result of their random motion
- Macroscopic view: migrating on a macroscopic scale from regions of high particle concentration into regions of low particles concentration.



Diffusion: Diffusion Current

□ Diffusion current

$$J_{p,diff} = -qD_p \nabla p$$

$$J_{n,diff} = qD_n \nabla n$$

where, D_p is hole diffusion coefficient, D_n is electron diffusion coefficient

□ Some discussions

- Difference in concentration: driving force of diffusion current.
- Both hole and electron will diffuse in the -x direction.

□ Total Current (including diffusion and drift of both electrons and holes)

$$\begin{aligned} J_p &= J_{p,drift} + J_{p,diff} = q\mu_p p E - qD_p \nabla p \\ &\quad \updownarrow \text{Drift} \quad \updownarrow \text{Diffusion} \\ J_n &= J_{n,drift} + J_{n,diff} = q\mu_n n E + qD_n \nabla n \end{aligned}$$

$$J = J_n + J_p$$

Diffusion: Einstein Relationship

- ❑ Relating Diffusion Coefficients/Mobilities (Einstein Relationship)

$$D_n/\mu_n = kT/q$$

$$D_p/\mu_p = kT/q$$

- ❑ Comments

- Derivation is omitted
- Valid even under non-equilibrium conditions, however the non-degenerate restriction still applies
- Typically value for Si and Ge at room temperature: $D/\mu \sim 0.026 \text{ V}$

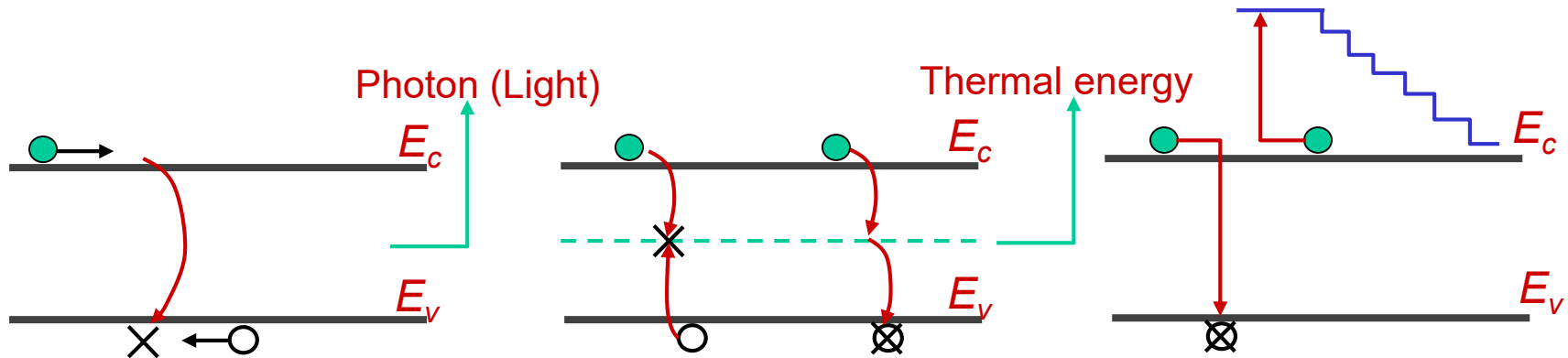
- ❑ Physical view: both reflect the ease of motion of carriers in crystal

	D_n (cm ² /s)	D_p (cm ² /s)	μ_n (cm ² /v-s)	μ_p (cm ² /v-s)
Ge	100	50	3900	1900
Si	35	12.5	1350	480

Recombination and Generation: Recombination

□ Carrier Recombination

- Definition: Recombination is a process whereby electrons and holes are annihilated or destroyed
- Three main recombination mechanisms
 - ✓ Band-to-band recombination
 - ✓ R-G center recombination
 - ✓ Auger recombination



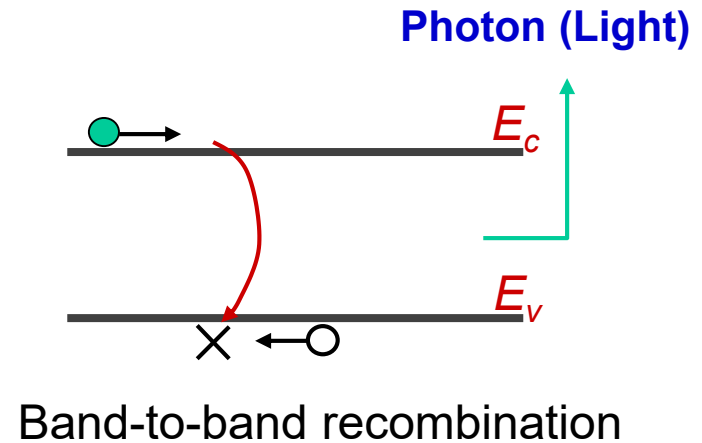
Band-to-band Recombination R-G center Recombination

Auger Recombination

Recombination: Band-to-Band Recombination

□ Band-to-band Recombination

- Simplest recombination process
- Involves the direct annihilation of a conduction band e^- and a valence band h^+
- The excess energy released during the process typically goes into the production of a photon (light)

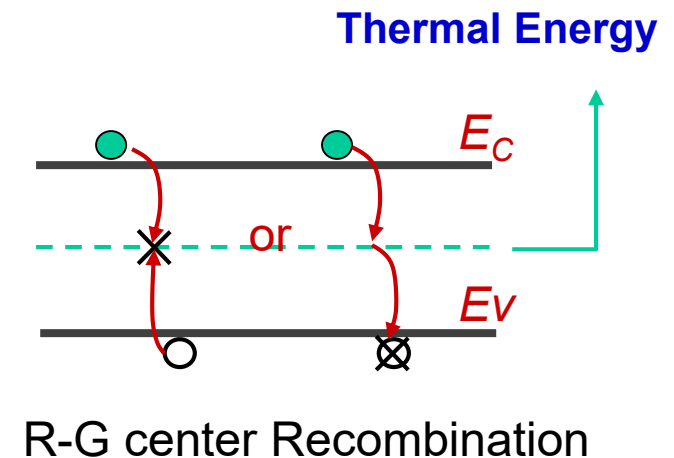


Recombination and Generation: R-G center

Recombination

□ R-G Center Recombination

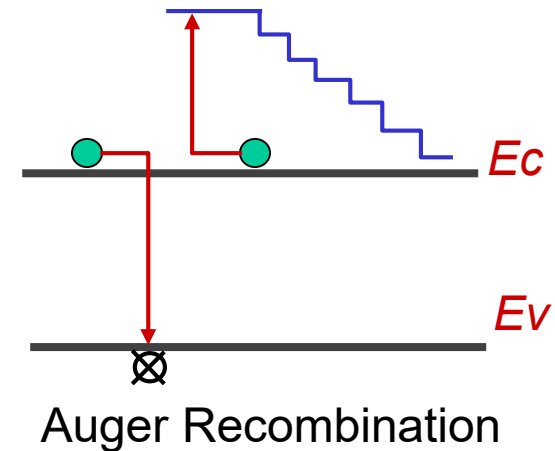
- Involves “third particle”
 - ✓ Takes place only at special locations within semiconductor known as R-G centers
 - ✓ R-G centers are lattice defects or special impurity atoms, such as Au in Si
- A two-step process
 - ✓ First, one type of carrier, say e^- , is trapped
 - ✓ Then, a hole is attracted to the trapped electron, or
 - ✓ The e^- lose energy again and annihilating the hole in the valence band
- Typically release thermal energy during the process or equivalent produce lattice vibrations



Recombination and Generation: Auger Recombination

□ Auger Recombination

- Three free carriers interact, two of the carriers recombine (band-to-band)
- The released energy is transferred into the surviving carrier (the third) during the collision
- The highly energetic carrier will lose energy in small steps through heat-producing collision with the semiconductor lattice

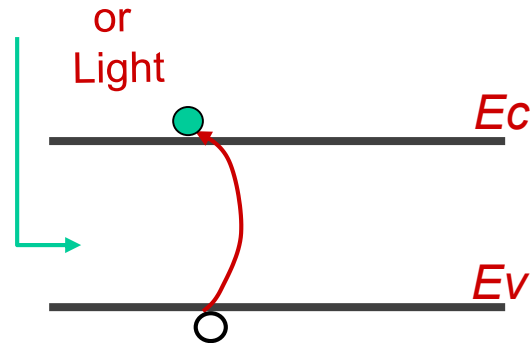


Recombination and Generation: Generation

□ Carrier Generation

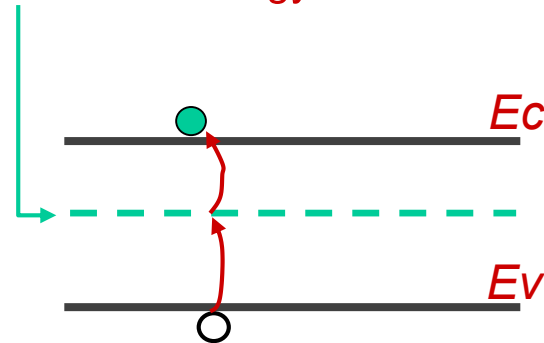
- Definition: Generation is a process whereby electrons and holes are created
- Three main generation mechanisms:
 - ✓ Band-to-band generation
 - ✓ R-G center generation
 - ✓ Carrier generation via impact ionization

Thermal Energy
or
Light

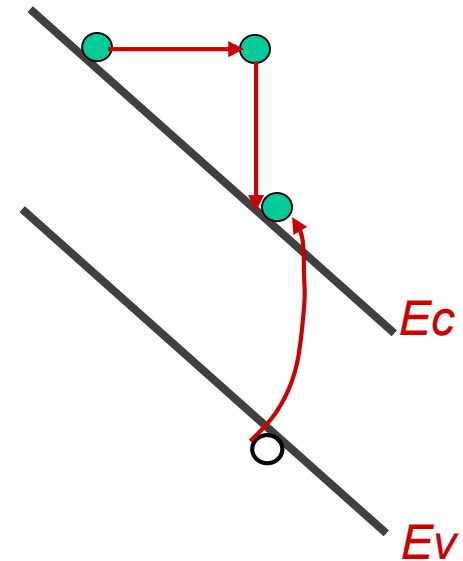


Band-to-band Generation

Thermal energy



R-G Center Generation



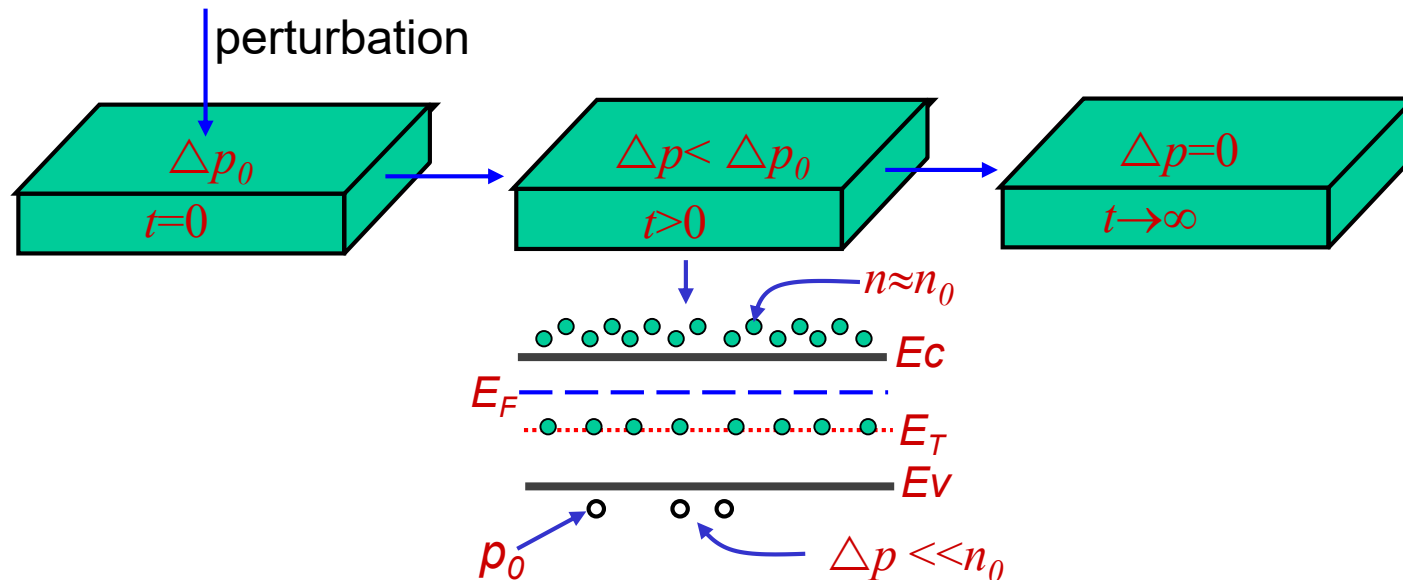
Carrier Generation
via Impact Ionization

Recombination and Generation: R-G Statistics

□ R-G Statistics

- Take indirect thermal recombination-generation as example
- Low-level injection

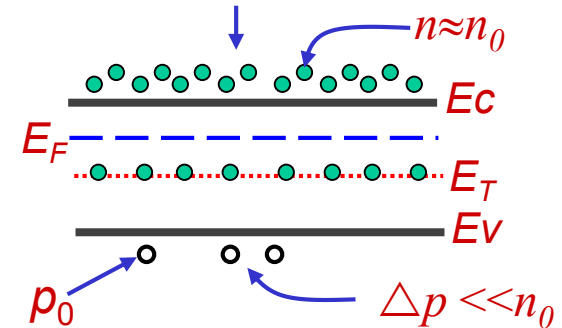
Low level injection implies
 $\Delta p \ll n_0$, $n \approx n_0$, in an n-type material
 $\Delta n \ll p_0$, $p \approx p_0$, in a p-type material



R-G Statistics: Hole Recombination Rate

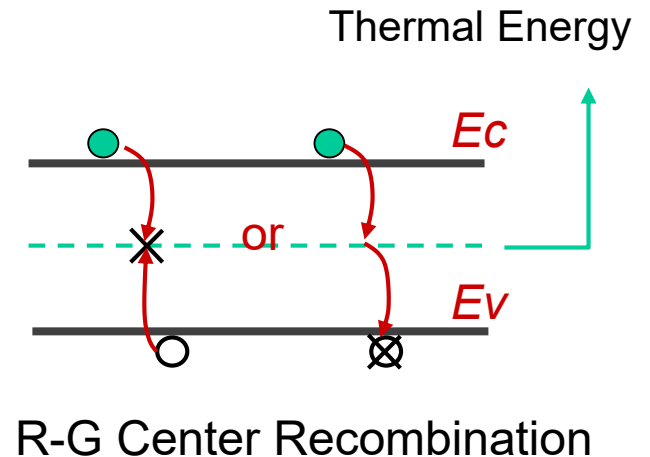
□ $\partial p / \partial t|_R$ (rate of hole recombination)

- Proportional to trap density (N_T) and hole concentration (p)
- $\partial p / \partial t|_R = -c_p N_T p$
- Where, c_p , capture coefficient, is a positive constant



Note:

- (1) Indirect recombination rate is proportional to the trap states occupied by electrons
- (2) Since n-type Si, E_T is below E_F , so almost all trap states are filled by electrons
- (3) Thus, the recombination rate is approximately proportional to trap density N_T



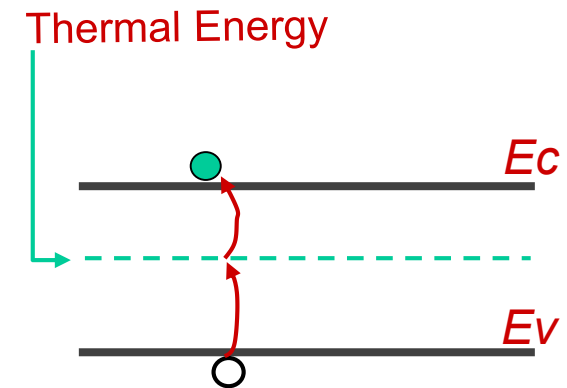
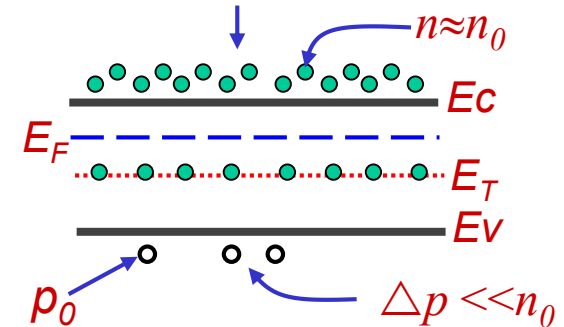
R-G Statistics: Hole Generation Rate

□ $\partial p / \partial t|_G$ (rate of hole generation)

- Depends only on the number of empty R-G centers (small for considering case)
- Number of empty centers is approximate constant at its equilibrium value
- $\partial p / \partial t|_G = \partial p / \partial t|_{G\text{-equilibrium}}$
 $= - \partial p / \partial t|_{R\text{-equilibrium}} = c_p N_T p_0$

Note:

- (1) Indirect generation rate is proportional to the trap states which are empty.
- (2) Since n-type Si, E_T is below E_F , so number of empty trap states is very small.
- (3) For same n-Si, number of empty trap states is also very small when there is no injection (in equilibrium).
- (4) Thus, generation rate is approximately proportional to empty trap density in equilibrium.



R-G Statistics: Net Change Rate of Holes

- Thus the net rate of change in the hole concentration due to the R-G center interaction is given by

$$\begin{aligned}\partial p / \partial t|_{\text{R-G}} &= \partial p / \partial t|_{\text{R}} + \partial p / \partial t|_{\text{G}} \\ &= -c_p N_T (p - p_0) \\ &= -c_p N_T \Delta p\end{aligned}$$

- Introduction of time constant: $\tau_p = (c_p N_T)^{-1}$ (discuss later)
- R-G center generation/recombination rate under low level injection

$$\partial p / \partial t|_{\text{R-G}} = -\Delta p / \tau_p$$

R-G Statistics: General Case

$$\left. \frac{\partial p}{\partial t} \right|_{\text{R-G}} = \left. \frac{\partial n}{\partial t} \right|_{\text{R-G}} = \frac{n_i^2 - np}{\tau_n(n+n_1) + \tau_p(p+p_1)}$$

where,
 $n_1 = n_i \exp[(E_T - E_i)/kT]$
 $p_1 = n_i \exp[(E_i - E_T)/kT]$

□ Comments:

- Valid for more general case (arbitrary injection levels and both carrier types in non-degenerate semiconductor).
- τ_n, τ_p , minority carriers lifetimes
 - ✓ Interpreted as average time an excess carrier can live in a sea of majority carriers
 - ✓ Depend on the R-G center concentration (N_T)
 - ✓ Control of lifetime (introducing of gold to increase N_T)
- low-level injection ($\Delta p \ll n_0$, n-type material)

$$\left. \frac{\partial p}{\partial t} \right|_{\text{R-G}} = -\frac{\Delta p}{\tau_p}$$

Equation of States: Continuity Equations

- Consider only hole current density with one dimension as example

$J_p(x+\Delta x)$: hole current leaving the volume $A \Delta x$

$J_p(x)$: hole current entering the volume $A \Delta x$

$\partial p / \partial t$: net increase in hole concentration per unit time,

- Relation: $\partial p / \partial t$ is the difference between the hole flux per unit volume entering and leaving, minus the recombination rate

$$\left. \frac{\partial p}{\partial t} \right|_{x \rightarrow x+\Delta x} = \frac{1}{q} \frac{J_p(x) - J_p(x+\Delta x)}{\Delta x} - \frac{\Delta p}{\tau_p}$$

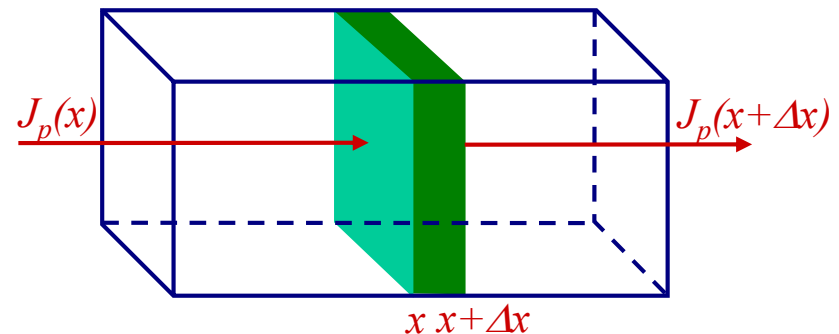
Rate of hole buildup
=
Increase of hole concentration in $A \Delta x$ per unit time
-
Recombination rate

- For $\Delta x \rightarrow 0$, we can get *continuity equation*

$$\frac{\partial p(x,t)}{\partial t} = \frac{\partial \Delta p}{\partial t} = -\frac{1}{q} \frac{\partial J_p}{\partial x} - \frac{\Delta p}{\tau_p}$$

- For electron minority carrier, similarly

$$\frac{\partial n(x,t)}{\partial t} = \frac{\partial \Delta n}{\partial t} = \frac{1}{q} \frac{\partial J_n}{\partial x} - \frac{\Delta n}{\tau_n}$$



Equation of States: Minority Carrier Diffusion Equations

- From continuity equation

$$\frac{\partial p(x,t)}{\partial t} = \frac{\partial \Delta p}{\partial t} = -\frac{1}{q} \frac{\partial J_p}{\partial x} - \frac{\Delta p}{\tau_p}$$

- From diffusion current definition

$$J_{p,diff} = -qD_p \frac{\partial p}{\partial x}$$

- Combine above 2 equations, we obtain

$$\boxed{\frac{\partial \Delta p}{\partial t} = D_p \frac{\partial^2 \Delta p}{\partial x^2} - \frac{\Delta p}{\tau_p}} \quad (\text{applying to holes in n-type material})$$

- For electron minority carriers, similarly

$$\boxed{\frac{\partial \Delta n}{\partial t} = D_n \frac{\partial^2 \Delta n}{\partial x^2} - \frac{\Delta n}{\tau_n}} \quad (\text{applying to electrons in p-type material})$$

- Often used in solving transient problems of diffusion with recombination

- Consider: what's assumption used in deriving above results? what will be the general form of equations?

Supplement Concepts: Diffusion Length

- In the steady state case, the diffusion equations simplified as:

$$\frac{\partial^2 \Delta p}{\partial x^2} = \frac{\Delta p}{D_p \tau_p} = \frac{\Delta p}{L_p^2}$$

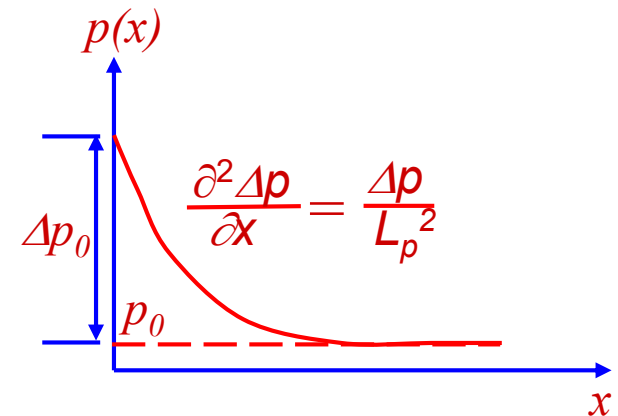
where, L_p is minority carrier diffusion length (hole in a n-type material)

- Definition:

$$L_p = \sqrt{D_p \tau_p}$$

- Physical view: represents the average distance minority carriers can diffuse into a sea of majority carriers before being annihilated
- Example: consider the steady state case of excess holes are injected into a semi-infinite semiconductor bar at $x=0$

$$\langle x \rangle = \frac{\int_0^\infty x \Delta p(x) dx}{\int_0^\infty \Delta p(x) dx} = L_p$$



$$p(x) = p_0 + \Delta p_0 \exp(-x/L_p)$$

$$\Delta p(x) = \Delta p_0 \exp(-x/L_p)$$

Supplement Concepts: Quasi-Fermi Level

- Quasi-fermi levels are energy levels used to specify the carrier concentrations inside semiconductor under non-equilibrium conditions.

- Definition:

$$\begin{array}{ll} n = n_i \exp(E_{Fn} - E_i) / kT & \text{or} \quad E_{Fn} = E_i + kT \ln(n / n_i) \\ p = n_i \exp(E_i - E_{Fp}) / kT & \text{or} \quad E_{Fp} = E_i - kT \ln(p / n_i) \end{array}$$

- Example:

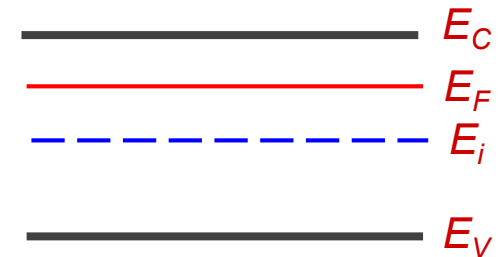
(a) $n_0 = N_D = 10^{15} \text{cm}^{-3}$ and $p_0 = 10^5 \text{cm}^{-3}$
(equilibrium)

(b) $p = p_0 + \Delta p = 10^{11} \text{cm}^{-3}$, $n \approx n_0 = 10^{15} \text{cm}^{-3}$
(nonequilibrium, steady state)

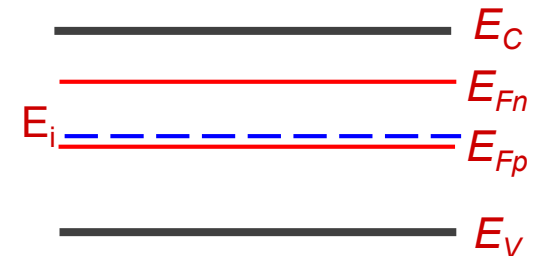
- Physical view:

$$J_p = \mu_p p \nabla E_{Fp}$$

namely, a quasi-Fermi level that varies with position
($\nabla E_{Fp} \neq 0$) indicates there is current flowing inside the semiconductor



Equilibrium



Non-equilibrium

Summary of Carrier Action

